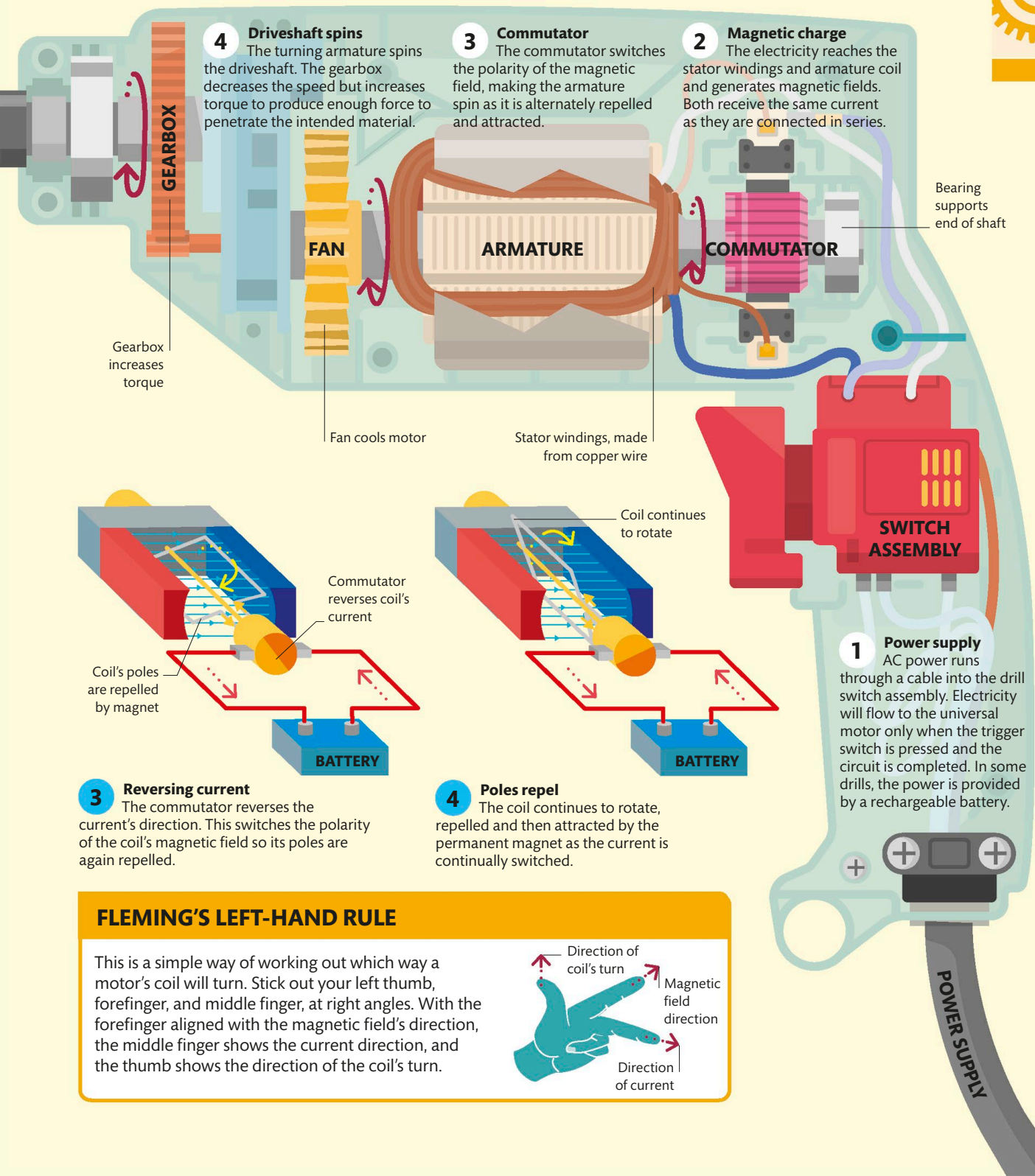


4 Driveshaft spins
The turning armature spins the driveshaft. The gearbox decreases the speed but increases torque to produce enough force to penetrate the intended material.

3 Commutator
The commutator switches the polarity of the magnetic field, making the armature spin as it is alternately repelled and attracted.

2 Magnetic charge
The electricity reaches the stator windings and armature coil and generates magnetic fields. Both receive the same current as they are connected in series.

1 Power supply
AC power runs through a cable into the drill switch assembly. Electricity will flow to the universal motor only when the trigger switch is pressed and the circuit is completed. In some drills, the power is provided by a rechargeable battery.

3 Reversing current
The commutator reverses the current's direction. This switches the polarity of the coil's magnetic field so its poles are again repelled.

4 Poles repel
The coil continues to rotate, repelled and then attracted by the permanent magnet as the current is continually switched.

FLEMING'S LEFT-HAND RULE

This is a simple way of working out which way a motor's coil will turn. Stick out your left thumb, forefinger, and middle finger, at right angles. With the forefinger aligned with the magnetic field's direction, the middle finger shows the current direction, and the thumb shows the direction of the coil's turn.